

Ethan Moon

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EDUCATION

Forest Hills Central High School

August 2022 – Present

- Cumulative GPA: (W) 4.342,
- SAT Superscore: 1520 (Math: 770, English: 750)
- AP Exams:
 - Science: AP Chemistry, AP Physics C: Mechanics, AP Physics C: Electricity & Magnetism, AP Environmental Science, AP Biology, AP Computer Science A,
 - Math: AP Precalculus, AP Calculus BC
 - Social Studies: AP U.S. History, AP World History, AP Macroeconomics, AP Microeconomics, AP World History,
 - English: AP Language & Composition, AP Literature & Composition
 - Language: AP Spanish Language & Composition,

Calvin University

- Dual Enrollment Student (Multivariable Calculus, Differential Equations with Linear Algebra)

University of Michigan (College of Engineering)

August 2026

RESEARCH/ENGINEERING EXPERIENCE

Bioprinting Researcher

October 2025-Present

Co-Author – Grand Valley State University (Mentor: Dr. Abishek Kamaraj)

- Co-authored and presented research at the ASEE North Central Conference on a closed-loop bioprinting system integrating high school, undergraduate, and graduate researchers
- Developed a CFD-based digital twin (OpenFOAM, interFoam) to model bio-ink extrusion under laminar, surface-tension-dominated flow
- Built a real-time feedback pipeline combining computer vision and sensor data to detect defects (e.g., clogging, filament thinning) and enable adaptive control
- Designed a machine vision classification system for filament quality and print accuracy using image-based analysis
- Improved system reliability by reducing false failure detection and enabling real-time correction during printing
- Collaborated in a multi-level mentorship model, coordinating across high school, undergraduate, and graduate researchers
- Preparing and expanding on current research for bioprinting journal publication
-  Draft_Paper_for_submission_R1.pdf
-  ASEE Mount Union Bioprinting Presentation (2).pdf

Grand Valley State University Hysteresis Research

Summer 2025

Lab assistant

- Researched methods to minimize aerodynamic hysteresis using low-cost techniques under Prof. Sanjivan Manoharan and graduate mentor Brendan Workman.
- Designed and 3D-printed airfoil prototypes for iterative testing.

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- Conducted wind tunnel experiments using Arduino-based sensors for real-time data acquisition.
- Analyzed flow behavior and hysteresis effects using CFD software and Excel.
- Focused on cost-effective alternatives to traditional hysteresis mitigation strategies.

Grand Valley State University Relative Motion of Satellites Research

Summer 2025

Researcher and Simulation Analyst

- Researched relative motion and orbital dynamics of satellites under the mentorship of Prof. Phillip Hittepop.
- Used Octave and NASA GMAT to simulate orbital transfers, relative trajectories, and control strategies.
- Coded and implemented orbital mechanics algorithms to analyze perturbations, rendezvous conditions, and satellite coordination.
- Interpreted simulation data to assess fuel-efficient maneuvers and stable formation flying.
- Explored practical applications in satellite swarms, autonomous navigation, and LEO constellation management.

Independent Engineering Project - “Healthy Eaters” (Self Directed)

- Designed and developed a sensor-based smart plate system integrated with a gamified mobile application to promote healthy eating behaviors in children with autism
- Built hardware prototype using load-cell weight sensors, microcontroller (Arduino/ESP32), and Bluetooth communication for real-time food intake tracking
- Developed a gamified app (Defold, Lua) with reward systems, personalized food recommendations, and progress tracking to encourage gradual dietary expansion
- Applied concepts from behavioral psychology and human-centered design to reduce food aversion and improve engagement through structured, low-pressure exposure
- Conducted real-world testing, demonstrating increased willingness to try new foods and improved dietary variety in users
- Integrated real-time data visualization and parental monitoring tools to enable data-driven feedback and intervention
-  Healthy Eaters (1).pdf

Internship, Casual Simulations

March 2024 – Present

Rowing System Simulation Optimization & Rowing App Sales

- Optimized performance and improved system functionality of rowing system simulations using XP exchange and Arduino technology, selling the rowing app to a high school rowing team.
- Applied problem-solving skills to enhance robotic systems, utilizing XP exchange and Arduino technology to optimize performance.
- Gained practical experience in simulation technologies, collaborating with professionals to develop advanced technical solutions and improve system functionality.

Western Michigan University Design Build Fly Team (WMU DBF)

January 2025 – Present

High School Participant

- Collaborated with university engineering students on the WMU Design Build Fly team to design, build, and test model aircraft for collegiate competitions.

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- Gained hands-on experience applying aerospace engineering principles to real-world flight systems.
- Contributed to performance optimization by assisting in design iteration, fabrication, and testing processes.

University of Illinois Urbana Champaign Aerospace Research Project

June 2024 – Present

Researcher/Published Author

- Developed and wrote a research paper through close collaboration with an experienced graduate student mentor at UIUC.
- Publishing research paper entitled “A Comparison of Different Alternative Fuels” in the National Journal of High School Science, which compared benefits, limitations, and suitability for replacing traditional kerosene in aviation.
-  Ethan_Moon_Research_Paper (2).docx

Advanced Math Instructor

December 2025-Present

Mathnasium

- Deliver personalized instruction in advanced mathematics (algebra, geometry, precalculus), adapting explanations to diverse learning styles and skill levels
- Mentor students to build problem-solving intuition and mathematical confidence, emphasizing conceptual understanding over rote memorization
- Diagnose learning gaps and design targeted practice plans, resulting in measurable improvement in student performance and engagement
- Communicate complex concepts clearly and efficiently, strengthening both technical communication and teaching skills

Michigan State University High School Engineering Institute Summer Program

Summer 2023

Participant

- Completed a one-week immersive program covering mechanical, aerospace, electrical, and environmental engineering.
- Built model bridges and energy-efficient structures, gained exposure to engineering software and programming, and engaged with faculty and industry professionals to learn about career pathways.

LEADERSHIP ACTIVITIES

Helping Hands, N.P.O.

February 2022 – Present

Founder/Head Coach

- Founded and manage a nationwide volunteer club providing soccer education to diverse and underserved youth.
- Recruited 10+ high-level and diverse high school coaches (ECNL/MLS Next) across the U.S..
- Built and maintain the official website ([helpinghandstraining.com](https://www.helpinghandstraining.com)); screen volunteer applications and essays to assign coaches to local programs based on need and fit.
- Raised over \$3,000+ in donations; featured in [Village 99's](#) media platforms ([IG](#)) for community impact.
- <https://www.helpinghandstraining.com/>

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Global Youth Mission STEM Club

April 2024 – Present

Leadership Cabinet (January 2025 – Present)

- Collaborate with fellow cabinet members to organize monthly meetings that explore various STEM fields and promote STEM education in the community.
- Lead STEM-related community service initiatives every month, engaging in hands-on activities and projects that support local organizations and STEM causes.
- <https://sites.google.com/view/gymstemclub/home>

Global Youth Mission

April 2024 – Present

Presidential Cabinet (January 2025 – Present)

- Coordinate monthly meetings and volunteer efforts including working with Zooniverse, sending letters to patients, and participating in other community service projects aimed at supporting individuals and causes in need.
- <https://www.globalyouthmission.org/index.aspx>

International STEM Research Journal

April 2024 – Present

Junior Editor (April - December 2024), Senior Editor (January 2025 – Present)

- Under mentorship of Ph.D Heather Killeen, produce short form editorials and provide copy-editing for biannual STEM publication.
- Provide constructive feedback and editorial support for high school STEM research papers submitted for publication.
- Contributed to the biannual publication of the journal, including Issue 4, which features a research paper entitled "Asian Carp are Ruining the Great Lakes." ([Link to Issue 4](#))

MLS NEXT Youth Soccer 2008 Player

August 2020 – June 2025

- Competed in the premier national youth soccer league (MLS NEXT) as the starting center back for five years, demonstrating elite-level performance in both regional and national tournaments.
- Served as Vice-Captain for the 2023–2024 season, leading the team through strategic decision-making and providing motivation on the field.
- Demonstrated exceptional time management by balancing rigorous training with academic responsibilities, maintaining performance excellence in both areas.

Forest Hills Central High School Varsity Soccer Team

August 2024 – Present

Captain, Defensive Midfielder, Central Defender

- Foster a culture of discipline and devise scoring strategies earning All-Conference, All-District, and All-Region 1st Teams, and All-State Honorable Mention honors in the 2024 season.
- Play a key role in team defense, leading strategies, motivating teammates, and fostering a culture of discipline and teamwork on and off the field.

Michigan Youth Leadership Camp (MYLEAD)

Summer 2024 – Summer 2025

Sole School Representative (Summer 2024), Volunteer Mentor (Summer 2025)

- Selected as the sole representative from my school to attend this esteemed leadership development program focused on leadership training, community service, and team-building exercises.
- Engaged in projects alongside a diverse group of peers and learned from inspirational speakers such as Dustin Dale and Anthony Tyus Jr..

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- Returned as a volunteer and alumni mentor in 2025, continuing to mentor and lead younger attendees through MYLEAD's program ([More about MYLEAD](#)).

FH Charitable Investments

August 2023 – Present

Senior Analyst (11th, 12th)

- Lead a team focused on facilitating impactful charitable investments and financial education for underserved communities.
- Manage strategic initiatives, ensuring alignment with organizational goals and social impact objectives.
- Collaborate with stakeholders to identify and execute investment opportunities that promote sustainable change.
- Provide mentorship and guidance to team members, fostering collaboration and accountability.

Math Tutor

2022 – Present

- Tutor K-12 students, primarily in the Korean community.
- Provide individualized support in arithmetic, Pre-Algebra, and Algebra 1 concepts to help improve academic performance.

EXTRACURRICULAR ACTIVITIES

National Honor Society

August 2024 – Present

Member (11th - Present)

- Maintained a high academic standing while actively participating in community service and leadership activities.
- Collaborated with peers on service projects aimed at benefiting local communities and fostering positive social impact.

Grand Rapids Public Museum (Planetarium)

April 2024 – Present

Volunteer

- Lead interactive tours for visitors in the Planetarium, explaining exhibits and educating the public on astronomy topics.
- Assisted with "Camp Curious" summer programs, supporting educational activities and creating engaging experiences for children.
- Contributed to event operations for special programs like Robotics Day and Arts and Crafts Day, facilitating hands-on learning and educational engagement.

Astro Photography Exhibit with Grand Rapids Planetarium

May 2025

Program/Exhibit Director

- Independent Project
- Astrophotography Outreach (Grand Rapids Public Museum)
- Initiated and proposed an astrophotography exhibit; contacted experts to source 25+ high-resolution images, which I donated to the museum. Project adopted and developed into an interactive discovery cart. I now present and explain the images to visitors, enhancing public engagement with astronomy.

Part-Time, Midwest United FC Soccer Club

September 2022 – Present

Operations Support

- Work in the office once a week, assisting with administrative tasks and providing visitor support.
- Assisted in organizing tournaments and tryouts, handling event logistics, registration, and coordination to ensure smooth execution.

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- Recorded men's soccer games for performance analysis and strategic planning, gaining experience in event management and game documentation.

Coaching and Mentoring Volunteer, Midwest United FC

August 2022 – Present

Future Stars Youth Soccer Program

- Coached youth players (grades 9–10), led training sessions, promoted teamwork, and helped maintain facilities through community clean-up efforts.

Certified Referee

May 2021 – Present

- Hold 6 years of certification and advanced-level qualification, officiating high-level competitive soccer matches, including NAL and ECNL games across various age groups.
- Continuously updated knowledge of the latest rules and regulations through workshops and training programs, ensuring consistent application of current standards.
- Paid job

COURSES AND PROGRAMS

FTE program in Vanderbilt University (Economics for Leaders)

Summer 2024

- Participated in the FTE(Foundation for Teaching Economics) program at Vanderbilt University during the summer, gaining in-depth knowledge of economics-related topics.

MITx

June – August 2024

Introduction to Aerospace Engineering (2 weeks)

- Completed a two-week intensive course on aerospace engineering fundamentals.
- Studied key topics such as flight principles, aerodynamics, propulsion systems, and spacecraft design. Developed foundational skills in orbital mechanics, flight dynamics, and aerodynamic analysis.

Autonomous Drone Racing Online Course | MIT Beaver Works Summer Institute

- Completed rigorous pre-admission course covering Python, ROS, SLAM, PID control, and computer vision. Developed autonomous drone navigation algorithms through simulation-based labs.

AI for Design and Optimization Course | Coursera (University of Michigan)

- Completed a project-based course on applying AI techniques such as genetic algorithms, neural networks, and reinforcement learning to engineering and design optimization using Python.
- <https://coursera.org/share/05844e7ab47b79e2d58ee59139dd018d>

IBM AI Engineering Professional Certificate | Coursera (IBM)

- Completed a multi-course program covering deep learning, machine learning, and AI deployment. Gained hands-on experience with neural networks, TensorFlow, PyTorch, and model deployment tools using Python.

HONORS AND AWARDS

ACADEMIC —

DECA State Champion, Integrated Marketing Campaign – Service | Michigan DECA — Feb 2026
Qualified to represent Michigan at the **International Career Development Conference (ICDC)**

-  *Service Integrated Market Campaign (13).pdf*

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2nd Place Overall & Grand Prize Honorable Mention, Southwestern Michigan Science & Engineering Fair — Mar 2025

Advanced to the **Michigan State Science & Engineering Fair**

Finalist, DECA District Competition — Sept 2024

Top performer in both **Roleplay and Written Exam**; qualified for State Competition

Dartmouth Book Award | Dartmouth College Alumni Association — 2025

Selected as **top junior** for exceptional academic achievement and leadership

AP Scholar with Distinction (11th), AP Scholar (10th) | College Board — 2024–2025

Recognized for sustained excellence across multiple AP exams

2nd Place, GYM International STEM Innovation Challenge — 2025

Finalist, International Astronomy & Astrophysics Competition (IAAC) — 2025

National Recognition Program Award Recipient | College Board — 2025

Distinction (Top 5%), International GYM Essay Contest — Summer 2024

Recognized for analytical writing on global societal challenges

Meijer Prep Athlete of the Week — 2025

Selected for athletic performance and leadership among all senior athletes in west michigan

VIOLIN —

Tokyo, Golden Classical Music Awards International Competition **2025**

First Prize Winner

- Invited to perform a solo at the prestigious Tokyo Opera City Hall as part of the international winner's recital.

Los Angeles, Golden Classical Music Awards International Competition **2024**

First Prize Winner

- Invited to Perform solo at Walt Disney Concert Hall in Los Angeles after international selection.

New York, Golden Classical International Music Awards **2024**

First Prize Winner

- Performed at Carnegie Hall.

The International Music Competition “Oxford” Grand Prize Virtuoso **2024**

First Prize Winner

- Invited to perform at the Sheldonian Theatre, University of Oxford.

VIVO (Winter) International Music Competition **2024-2025**

Second Place

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- Performer at Winners' Concert, Merkin Concert Hall, New York

Protege International Music Talent Competition

2024

Second Place

- Recognized for superior talent and artistic expression, with a performance opportunity at Carnegie Hall.

Great Composers Competition: The Music of the Centuries

Second Place – Violin (2024)

- Awarded for exceptional interpretation and technical mastery in a historical music category.

Stulberg International String Competition winner

2024

Master Class Participant

- Selected to join an elite master class, receiving advanced training from world-renowned violinist: ANNALEE PATIPATANAKOON.

MSBOA State Solo and Ensemble Competition

2023

Michigan State Award Recipient, Blue Ribbon Recipient

- Received a Superior rating, Blue Ribbon for exceptional performance in solo and ensemble categories, reflecting high-level musical proficiency and artistry.

SKILLS

- **Programming & Software:** Python, Java, MATLAB/Octave, Arduino
- **Engineering & Simulation:** Computational Fluid Dynamics (CFD), OpenFOAM, NASA GMAT, OCTAVE, CAD & 3D Printing, LASER CUTTING
- **AI & Data:** Machine Learning fundamentals, anomaly detection, computer vision basics, data analysis
- **Research & Technical:** Experimental design, scientific writing, academic publishing, technical presentations